Differentiating and Integrating Measured Motion

Model-free estimation of velocity and acceleration from displacement, and of velocity and displacement from acceleration

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Two problems every vibration lab has

The setting: one sensor type, and you need the other kinematic quantities

- You measured displacement (laser vibrometer, DIC, encoder) and you need velocity and acceleration.

  Restoring-force surfaces, NNM energy balance, damping identification, model updating.

- You measured acceleration (the cheap, rugged, ubiquitous sensor) and you need velocity and displacement.

  Mode shapes in physical units, drift-free response, base-excitation compensation, fatigue cycles.

- Both are routine requests. Both are harder than they look.

- The naive answers — finite differences one way, cumulative trapezoid the other — fail, and fail differently.

They are the same problem, run in opposite directions

One framing that organises the whole landscape

- Write the measurement as $y = A u + e$, with A the integration operator.

  Differentiation: u is the derivative you want, A integrates it back to what you measured.

  Integration: u is the displacement, A integrates the acceleration twice.

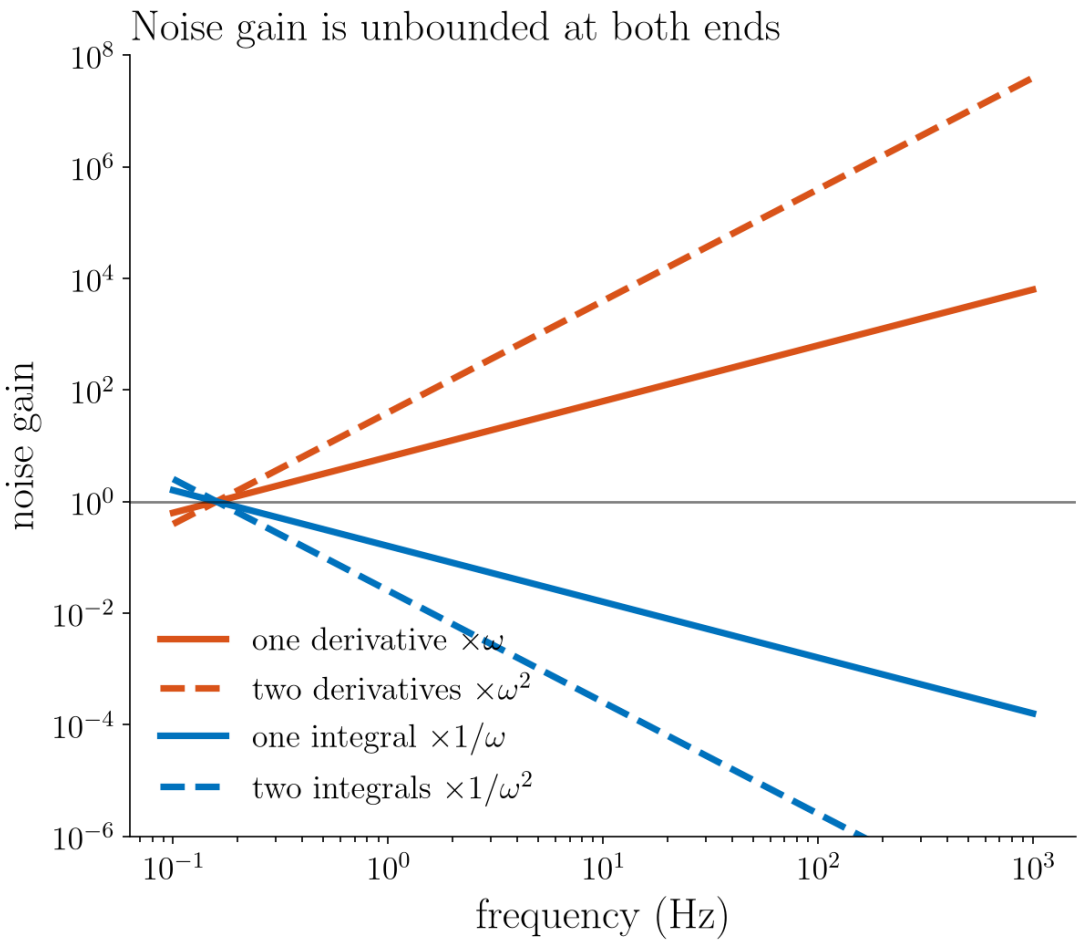
- A is compact. Its inverse is unbounded. This is the definition of an ill-posed inverse problem.

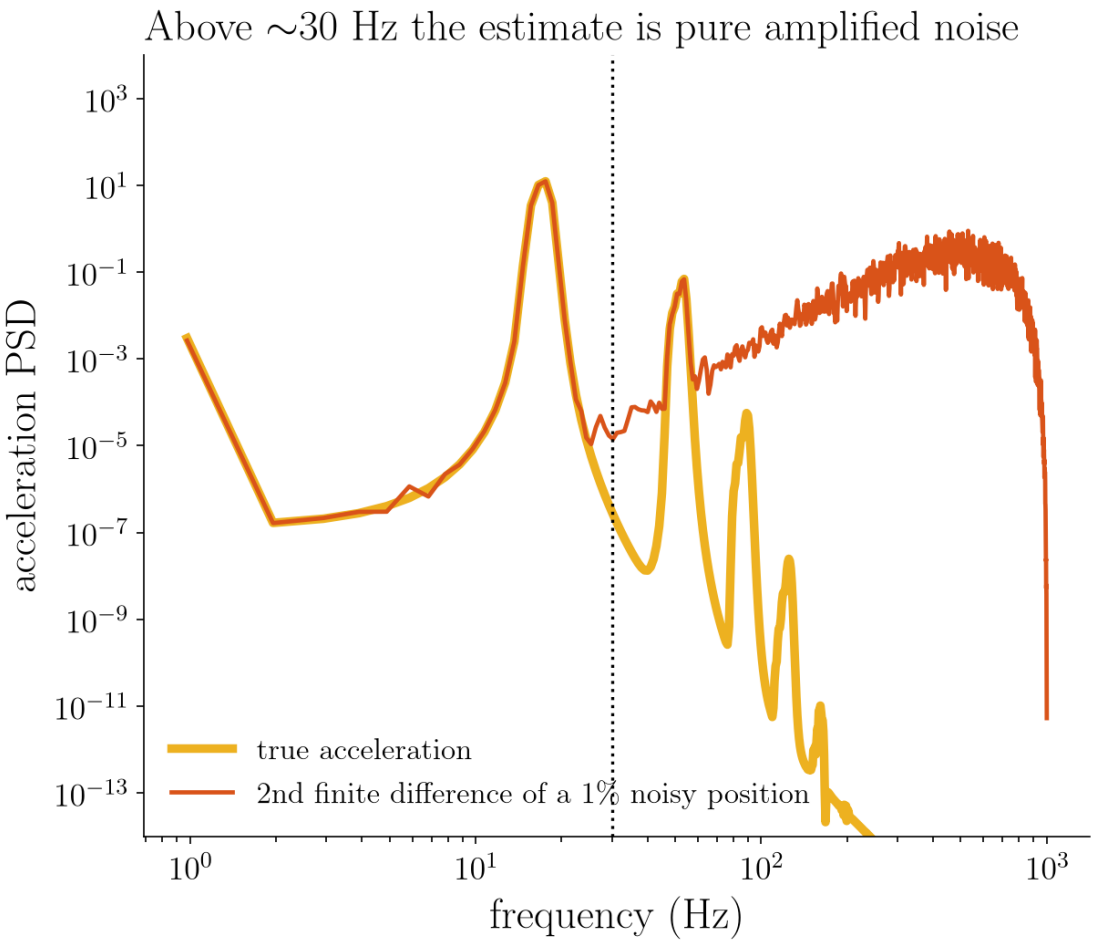
- Noise is amplified without limit — by ω per derivative going up, by 1/ω per integration going down.

- No amount of cleverness removes this. It is a property of the operator, not of your algorithm.

- Every usable method therefore adds a regularity prior. That is the only thing that distinguishes them.

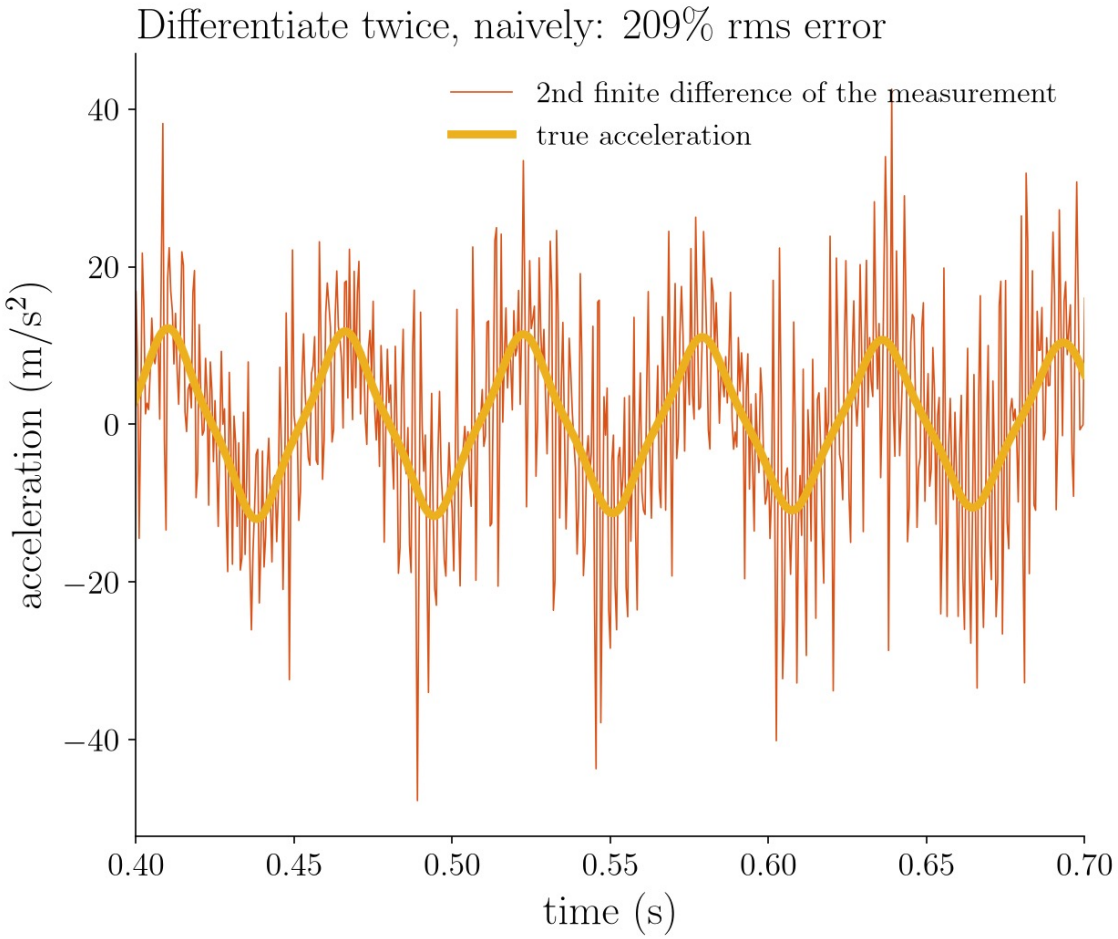
Why it is ill-posed

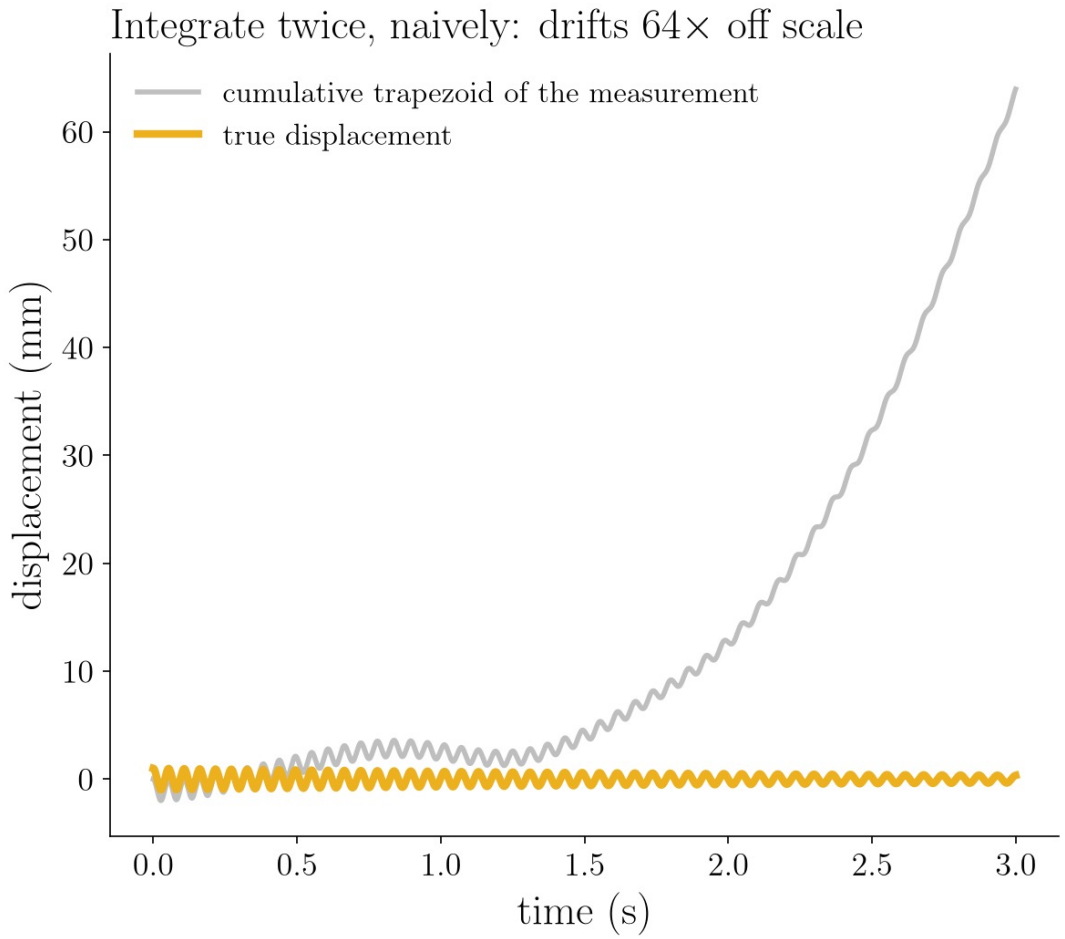




Left: the amplification factor is unbounded at both ends. Right: the same effect on a real record — a 1% noise floor on position dominates the acceleration estimate above about 30 Hz.

What happens if you ignore it





Left: second central difference of a 1% noisy position — 209% rms error. Right: plain cumulative trapezoid of an accelerometer with a 0.5% offset — 64x off scale in 3 s.

The only cure: state a prior

Methods differ only in which regularity prior they impose — pick the row you actually believe

What you are willing to assume	The method it gives you
The signal is bandlimited and periodic	Spectral / FFT differentiation
It is locally a low-order polynomial	Savitzky–Golay, Lanczos
It lies in a low-rank delay-embedding subspace	SVD / SSA
Its p-th derivative is small in L2	Whittaker–Henderson, Tikhonov, smoothing spline
Its derivative is piecewise constant (L1)	Total variation (Chartrand)
It is an integrated Wiener process	Kalman / RTS smoother ≡ Gaussian process
Its derivative is bounded by a known constant	Levant sliding-mode (causal, online)

Spectral (FFT) differentiation

Multiply the spectrum by $(i\omega)^n$, roll off above a cutoff, transform back.

STRENGTHS

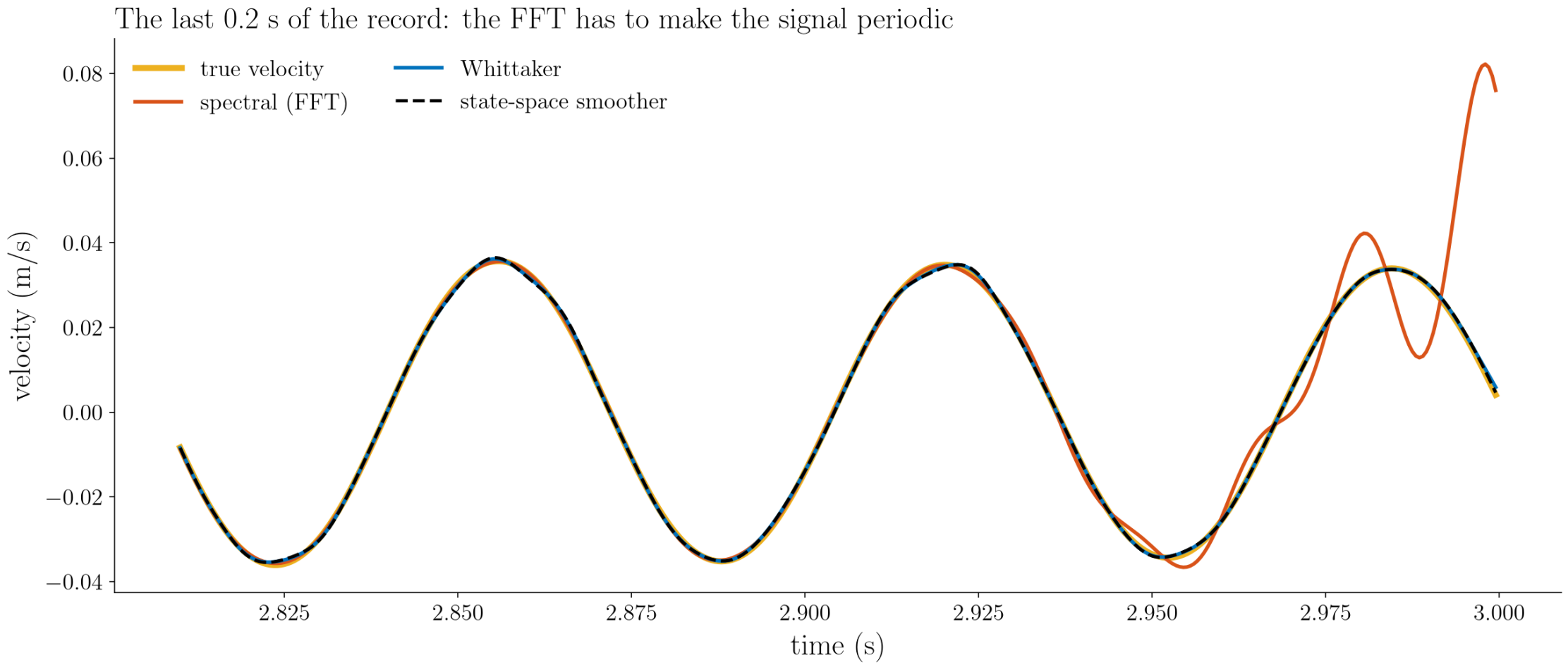
- Exact for a bandlimited periodic signal — no discretisation error at all
- Most accurate of any method in the interior of a long record (0.54% velocity error here)
- Trivial to implement, O(N log N), and the cutoff has an obvious physical meaning
- The natural choice when you already work in the frequency domain

LIMITATIONS

- Enforces periodicity — catastrophic at the record ends (next slide)
- Needs uniform sampling and a gap-free record
- Cutoff must be chosen; too high amplifies noise, too low removes signal
- Leakage if the record is not an integer number of periods (taper, then divide it out)

USE IT WHEN The record is long, uniformly sampled, and you only care about the middle of it.

The FFT caveat that costs people real results



Velocity over the last 0.2 s of a 3 s free decay. Interior error 0.54%; over the full record, 8.4% — a 15x degradation.

Savitzky–Golay

Fit a low-order polynomial in a sliding window; differentiate the polynomial analytically.

STRENGTHS

- Simple, fast, and available in every toolbox
- No periodicity assumption — much better behaved at the ends than the FFT
- Preserves peak heights and widths better than a plain moving average
- Window length maps to a cutoff via Schafer's rule, so it can be set from physics

LIMITATIONS

- Two knobs (window, polynomial order) that interact
- Implicitly assumes the signal is polynomial over the window — poor for lightly damped resonance
- Noticeably worse than an L2 penalty for second derivatives (2.4% vs 0.9% here)
- Edge behaviour depends on the padding convention, which differs between implementations

USE IT WHEN You want something defensible in ten lines with no linear solve, and first derivatives are enough.

Whittaker–Henderson and Tikhonov

Penalise the p-th derivative in L2: min ‖s − x‖² + λ‖Dᵖs‖². One banded solve.

STRENGTHS

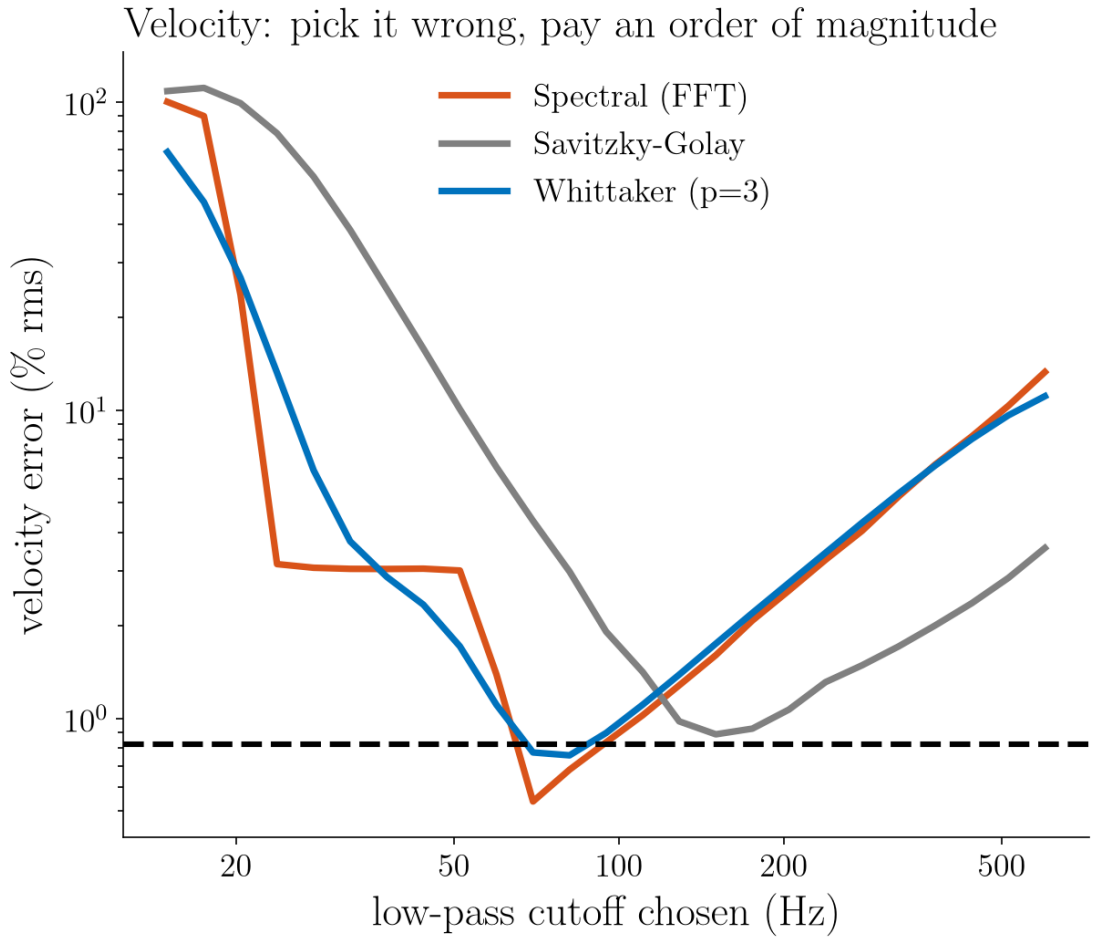
- One sparse banded solve — O(N), no iteration, no convergence to check
- Flat accuracy across the whole record, ends included
- λ has a closed-form map to a cutoff frequency, so no blind parameter sweep
- Same machinery runs the integration direction (Tikhonov) — one implementation, both tasks

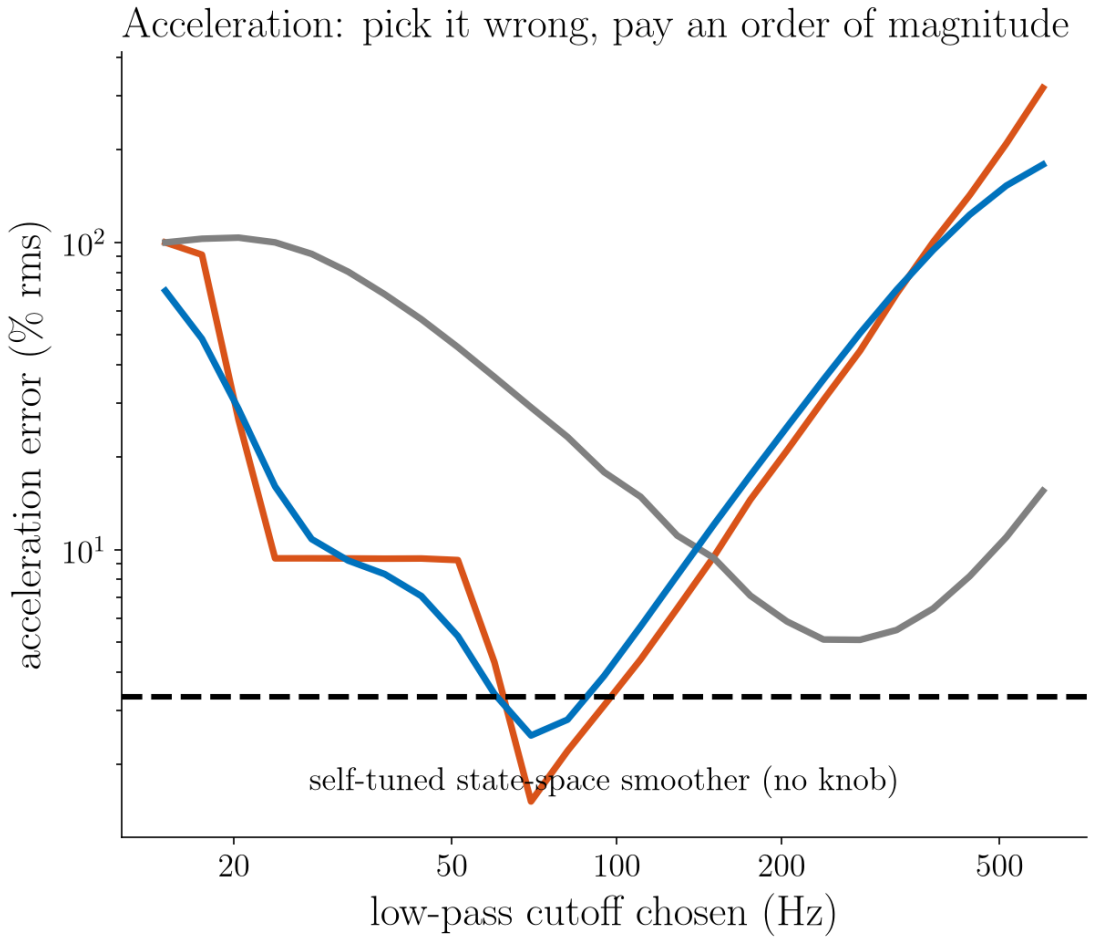
LIMITATIONS

- Still one knob, and the optimum is sharp for second derivatives
- Quadratic penalty rounds off genuine corners — wrong prior for impacting systems
- The normal matrix is badly scaled; use a sparse direct solver, not an iterative one
- No uncertainty estimate comes out of it

USE IT WHEN This is the sensible default for smooth vibration data. Start here.

Every method has an optimum you have to find





Error against exact ground truth as the cutoff is swept, 1% position noise. The dashed line is the self-tuned state-space smoother, which has no knob at all.

Setting the knob from physics, not from a sweep

Every L2 method has a closed-form map from a corner frequency in Hz to its penalty weight

Method	Parameter	Rule	Set it from
Whittaker–Henderson (order p)	λ	$\lambda = (2\pi f_c)^{-2p}$	Highest mode you care about
Tikhonov integration (order n)	λ	$\lambda = (2\pi f_c)^{2n}$	Lowest frequency you trust
Savitzky–Golay	window 2M+1	$f_c \Delta t \approx (k+1)/(3.2M - 4.6)$	Schafer's –3 dB rule
Spectral	cutoff	direct	Where the PSD meets the noise floor
Total variation	α	no closed form	Sweep; scale-sensitive, normalise first
State-space smoother	q	fitted automatically	Marginal likelihood — nothing to set

Total variation (L1)

Penalise $\|Du\|_1$ instead of $\|Du\|_2$. The L1 norm does not punish a single large jump.

STRENGTHS

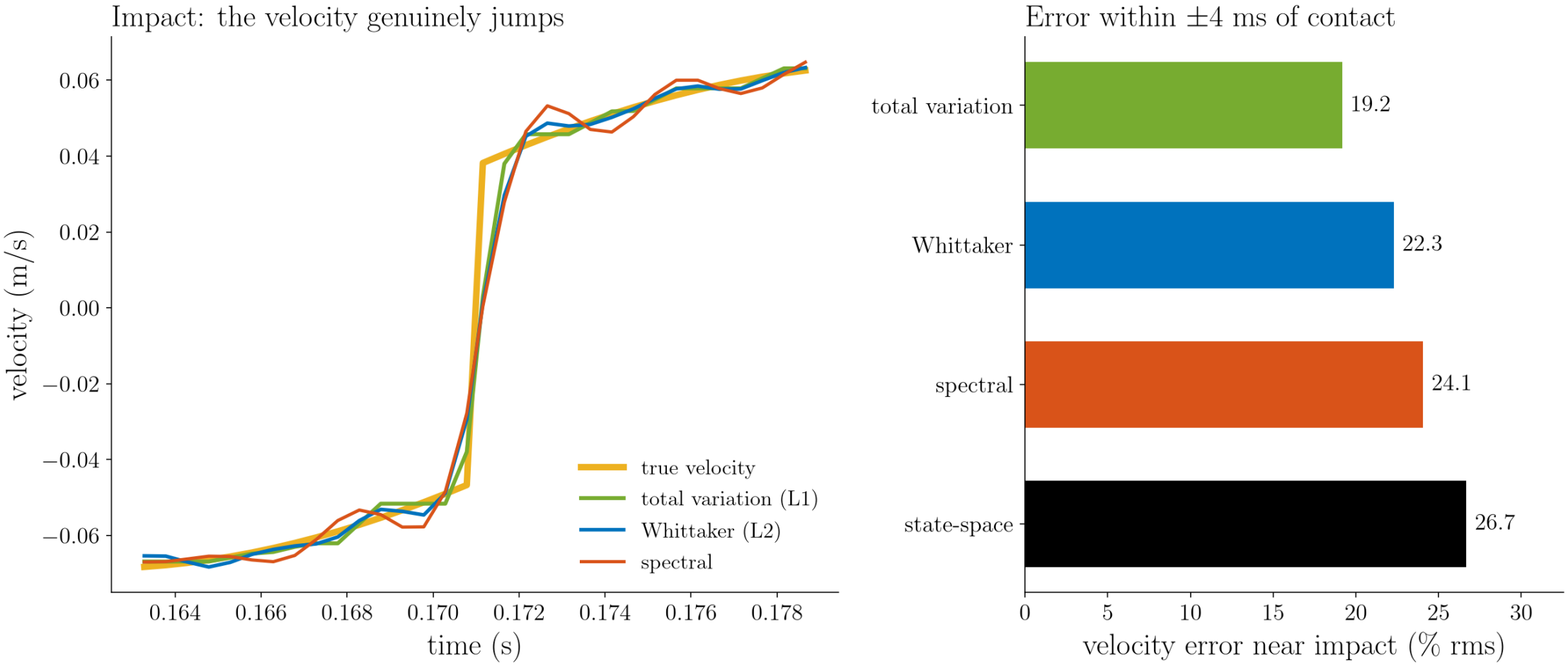
- The only family that preserves a genuine discontinuity in the derivative
- Best method tested at an impact: 19.2% vs 22–27% for every quadratic penalty
- Physically the right prior for impacts, stick–slip, backlash, dry friction reversal
- Recovers a piecewise-constant derivative exactly in the noise-free limit

LIMITATIONS

- Several times worse than an L2 penalty on smooth signals — 5.1% vs 0.5% here
- Iterative, no closed form, and the objective is non-smooth
- α is tied to absolute signal scale — normalise before, rescale after
- The naive solver silently fails to converge (next slide)

USE IT WHEN Your derivative genuinely jumps. Otherwise do not — it is the wrong prior for smooth data.

Where the L1 penalty earns its place



Vibro-impact oscillator, restitution 0.85. Left: one impact, velocity jumps. Right: velocity error within ±4 ms of contact, best setting per method.

A cautionary tale: the solver was the result

Reported numbers for an iterative method are claims about the solver as much as the estimator

- Chartrand's formulation has a dense, severely ill-conditioned Hessian $A^TA + \alpha L$.
- Conjugate gradients stalls at any iteration cap. IRLS on the smoothed L1 stalls too — measured rate 0.997 per iteration.
- The unconverged iterate acts as extra smoothing and silently changes the answer.

  MATLAB's pcg warns. scipy's cg returns the same unconverged vector with no warning at all.
- Three successive solvers moved the same benchmark:

  smooth signal 3.77% → 4.93% → 5.07% near impacts 17.9% → 16.6% → 19.2%
- Under-convergence had been flattering the method in both directions. Only the last pair is the true minimiser.
- Fix: substitute w = Au to get L1 trend filtering — the ADMM system is pentadiagonal and constant, so it factors once.

State-space: the Kalman / RTS kinematic smoother

Model the signal as an integrated random walk; state = [x, v, a, jerk, ...]. Smooth, do not filter.

STRENGTHS

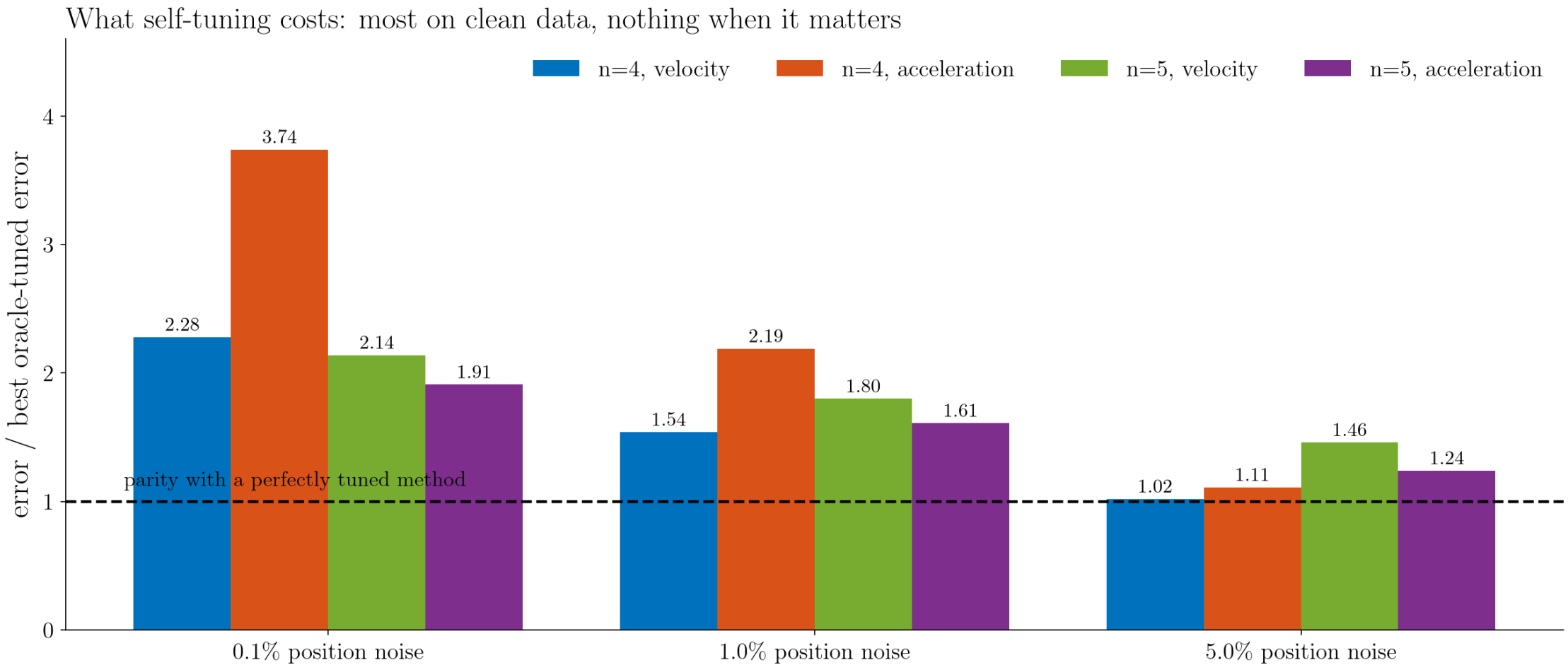
- Self-tuning — the single parameter q is fitted by marginal likelihood, no sweep
- Handles gaps, non-uniform sampling and multirate channels natively (NaN = missing)
- Returns a posterior standard deviation — error bars, not just an estimate
- Does differentiation, integration AND sensor fusion with one implementation
- O(N), and equivalent to a Gaussian process without the $O(N^3)$

LIMITATIONS

- Costs something against a perfectly tuned method on very clean data (up to 3.7x)
- The prior is smoothness — it rounds corners just as the L2 penalties do
- More code than a banded solve; the likelihood optimisation needs a sane bracket
- Bias states are weakly identified over short records — a drift absorber, not a calibration

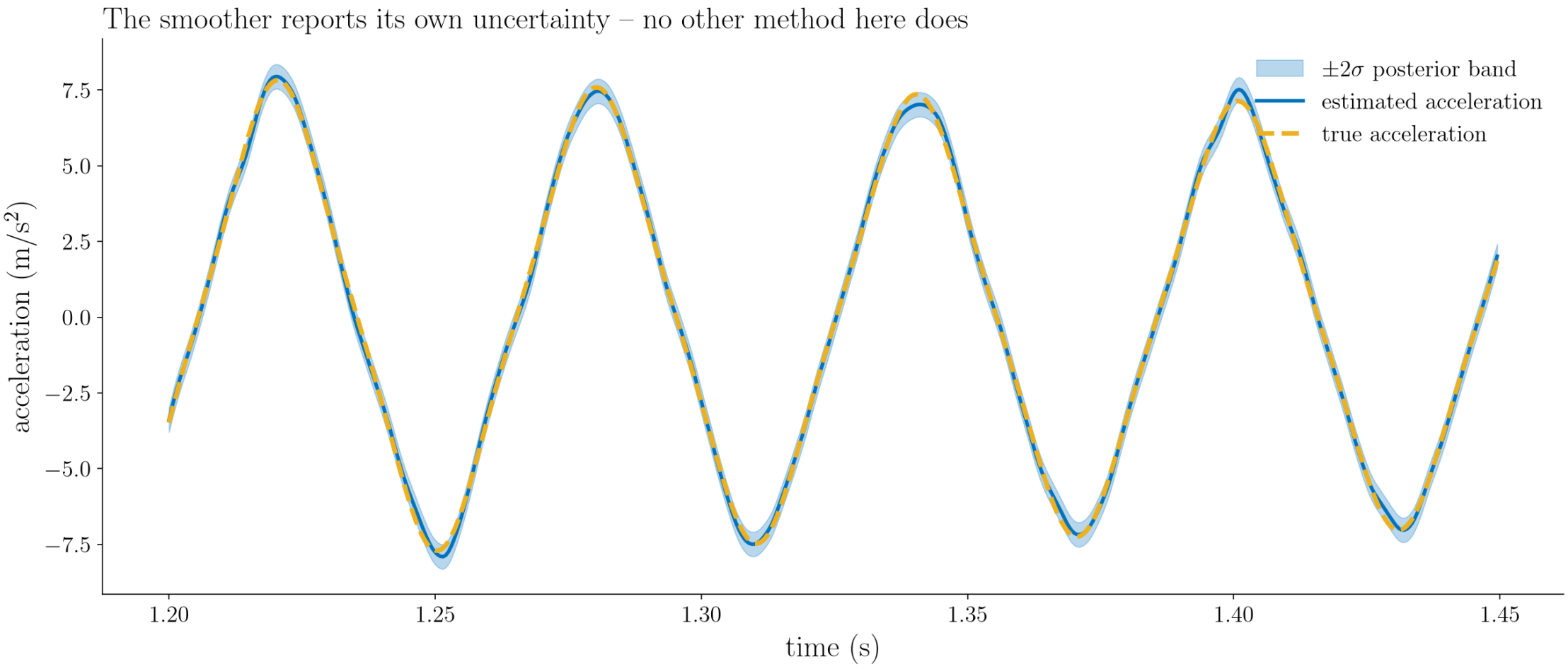
USE IT WHEN You want one tool that covers every case, or you have two sensors, or you need uncertainty.

What does self-tuning actually cost?



Ratio of the self-tuned error to the best error achieved by ANY method in this study with its knob tuned against ground truth.

It tells you how much to trust it



Acceleration from a noisy position record with its ±2σ posterior band. No other method in this comparison produces one.

Integration has an extra problem: the constants

Two integration constants are unobservable — every method makes an assumption, so make yours explicit

- Acceleration determines displacement only up to $x_0 + v_0 t$. Those two constants are not in the data.
- A DC offset of 0.5% integrates twice into a parabola that swamps the signal within seconds.
- Every working method therefore imposes a high-pass — the question is whether it admits to it.

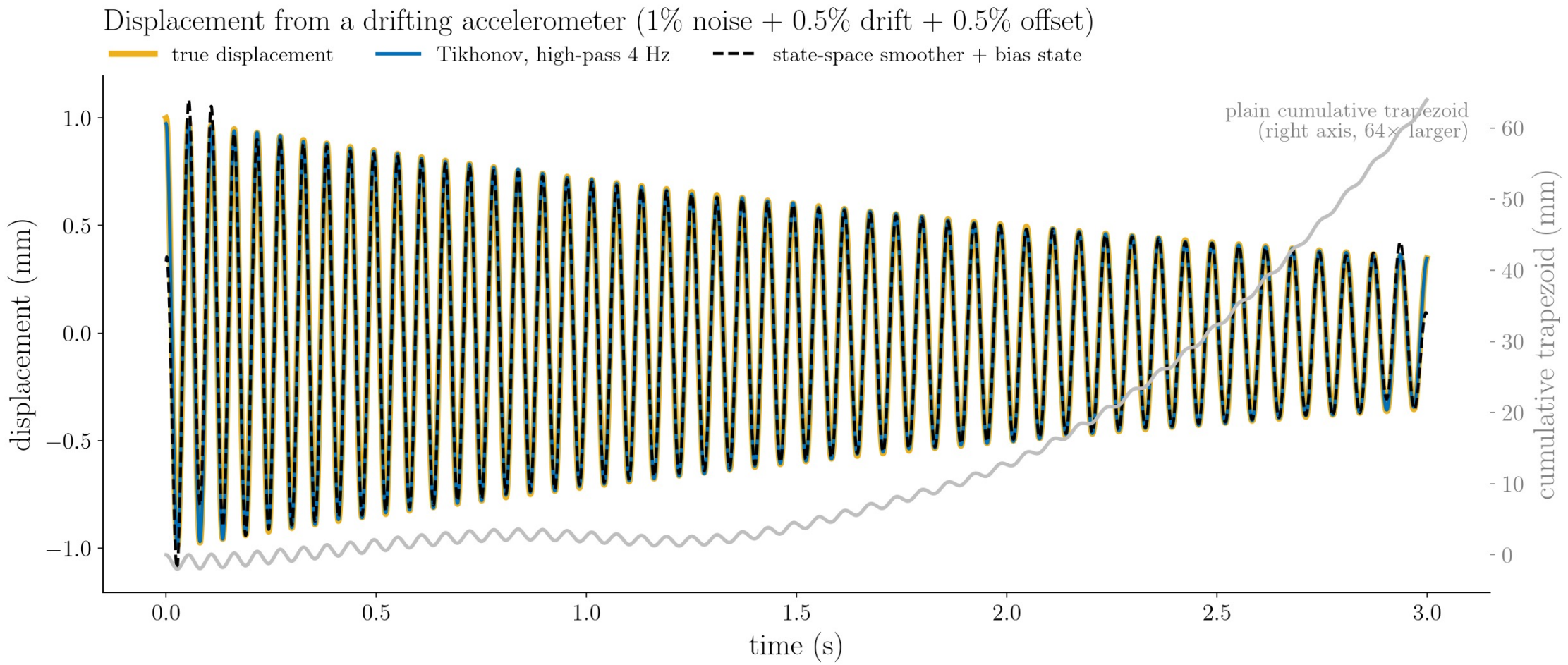
  Detrend + high-pass filter: fast, but the corner is chosen by eye and distorts the low modes.

  Tikhonov: the corner appears explicitly as $\lambda = (2\pi f_c)^{2n}$. One banded solve.

  State-space with a bias state: the corner becomes a prior on the unobservable directions.

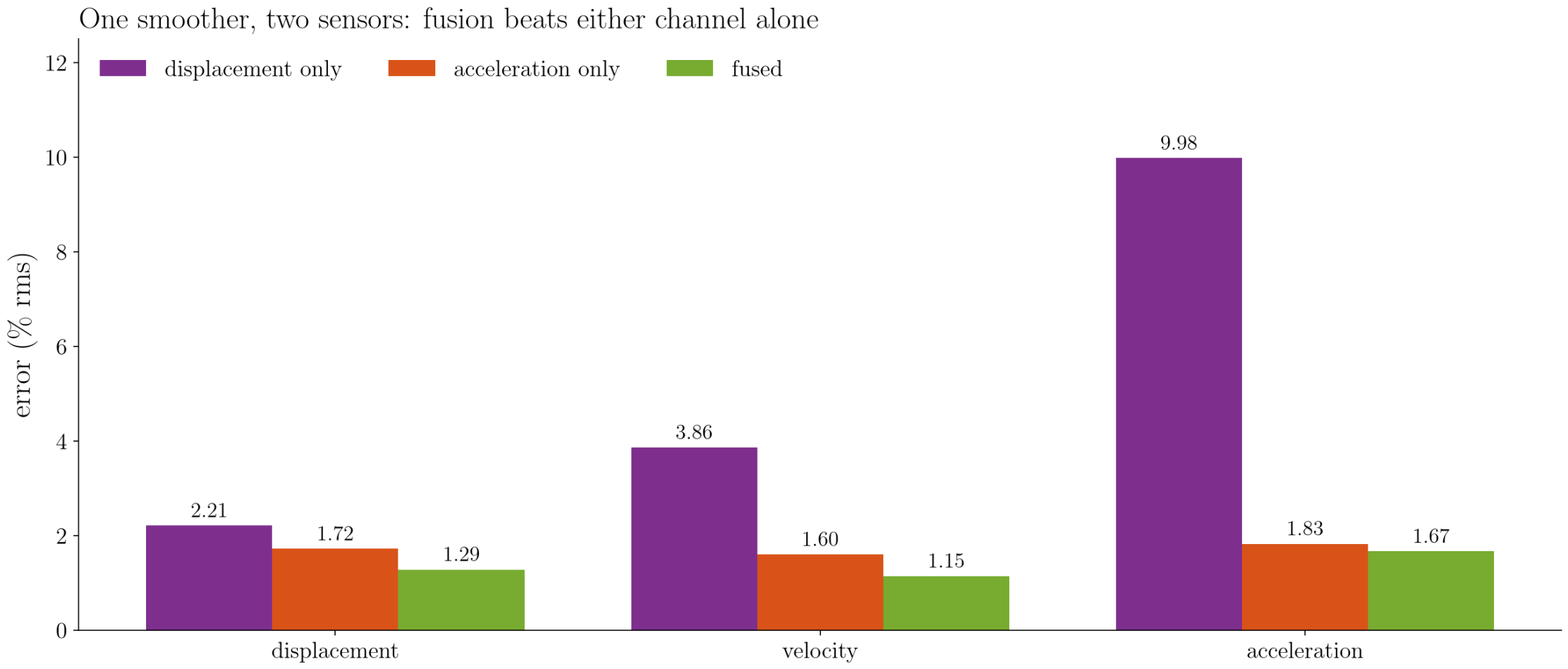
- Choose the corner below your lowest mode of interest and above the drift band — and say what you chose.
- If you also have a displacement channel, you do not need the corner at all. That is the next slide.

Integration done three ways



Displacement recovered from an accelerometer with 1% noise, 0.5% drift and a 0.5% offset. Regularised methods overlay the truth; the plain trapezoid needs its own axis.

If you have both sensors, use both at once



One state-space smoother, 200 Hz displacement at 3% noise fused with 2 kHz acceleration at 1% noise plus drift.

The comparison on one slide

Knobs = free parameters you must choose; the state-space smoother fits its one parameter itself

Method	Prior	Knobs	Ends	Gaps	Error bars	Best at
Spectral (FFT)	bandlimited	1	poor	no	no	long uniform records
Savitzky–Golay	local polynomial	2	fair	no	no	quick first derivatives
Whittaker / Tikhonov	small p-th deriv.	1	good	no	no	smooth data, both directions
Smoothing spline	small p-th deriv.	1	good	yes	no	non-uniform sampling
Total variation	piecewise-const.	1	good	no	no	impacts, stick–slip
State-space smoother	integrated Wiener	0	good	yes	yes	everything else, and fusion
Levant sliding mode	bounded deriv.	2	n/a	no	no	online / control loops

What to actually do

A decision list, in the order the questions actually arise

- Smooth data, one sensor, want it simple → Whittaker–Henderson, cutoff from physics.
- Smooth data, want no knob or need uncertainty → state-space smoother, nstate = 4 (5 if very clean).
- Two sensors → always fuse them in one smoother. Never process them separately and reconcile after.
- Impacts, backlash, stick–slip → total variation, and check that the solver converged.
- Integrating acceleration alone → Tikhonov or a bias-state smoother; report the high-pass corner.
- Online or in a control loop → Levant's differentiator; everything else here is offline and zero-phase.
- Never: a bare finite difference, or a bare cumulative trapezoid.

Validating on a record with no ground truth

Four checks that cost minutes and one habit that costs nothing

- Everything shown here was measured against an exact solution. On your data you have none — so:
- Cross-check. Differentiate the estimated displacement twice; compare with the measured acceleration in the band you trust.
- Read the posterior band. If the feature you are interpreting is narrower than ±2σ, it is not there.
- Move the knob. Change the cutoff by a factor of two either way; if the answer moves more than the band, report that.
- Hold out a channel. With two sensors, fit on one and predict the other — the honest test.
- State your choices. Method, cutoff or high-pass corner, and solver convergence, alongside the sampling rate.

Summary

- Differentiating displacement and integrating acceleration are one ill-posed problem, not two.
- Noise amplification is a property of the operator. Every method you can choose adds a prior instead.
- So the real question is never 'which filter' — it is 'what do I believe about this signal?'
- For smooth vibration data: a quadratic penalty, tuned from a cutoff in hertz.
- For a genuine corner: an L1 penalty, with the solver's convergence checked.
- For anything else — gaps, two sensors, no time to tune, or a need for error bars: the state-space smoother.
- And if you have displacement and acceleration, fuse them. It is strictly better and it removes a knob.

Implementation and references

- Python and base-MATLAB implementations of every method here, with identical function names.

  kin_diff_whittaker, kin_diff_spectral, kin_diff_savgol, kin_diff_tvreg

  kin_int_tikhonov, kin_int_spectral, kin_noise_var

  kin_smooth — differentiation, integration and fusion in one self-tuning routine

- Cross-validated between the two languages: 12 of 12 checks pass, agreement 1e-16 to 1e-5.
- Key references:

  Chartrand (2011), Numerical differentiation of noisy, nonsmooth data — ISRN Appl. Math.

  Eilers (2003), A perfect smoother — Anal. Chem. 75:3631 (Whittaker–Henderson)

  Särkkä & Solin, Applied Stochastic Differential Equations, CUP 2019 (state-space / GP equivalence)

  Schafer (2011), What is a Savitzky–Golay filter? — IEEE Signal Process. Mag. 28(4):111

  Kim, Koh, Boyd & Gorinevsky (2009), l1 trend filtering — SIAM Review 51(2):339

  Levant (1998), Robust exact differentiation via sliding mode technique — Automatica 34:379